

SHETH L.U.J AND SIR M.V COLLEGE

Practical 2 A* Search and Recursive Best-First Search

CODE:

```
import heapq
import networkx as nx
import matplotlib.pyplot as plt
```

```
graph = {
    'Andheri Railway Station': {'DN Nagar Metro': 3, 'Jogeshwari': 4},
    'DN Nagar Metro': {'Lokhandwala Circle': 2, 'Versova Metro': 3},
    'Jogeshwari': {'Lokhandwala Circle': 5},
    'Lokhandwala Circle': {'Infinity Mall': 2, 'Versova Metro': 3},
    'Versova Metro': {'Infinity Mall': 2},
    'Infinity Mall': {}
}
```

```
h = {
    'Andheri Railway Station': 8,
    'DN Nagar Metro': 5,
    'Jogeshwari': 6,
    'Lokhandwala Circle': 2,
    'Versova Metro': 3,
    'Infinity Mall': 0
}
```

```
pos = {
    'Andheri Railway Station': (2,8),
    'DN Nagar Metro': (1,6),
    'Jogeshwari': (3,6),
    'Lokhandwala Circle': (1,4),
    'Versova Metro': (3,4),
    'Infinity Mall': (2,2)
}
```

```
def astar(start, goal):
    pq = [(h[start], start, [start], 0)]
    visited = set()

    while pq:
        f, node, path, g = heapq.heappop(pq)

        if node in visited:
            continue

        if node == goal:
            return path, g

        visited.add(node)

        for nxt, cost in graph[node].items():
```

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```
        if nxt not in visited:
            ng = g + cost
            heapq.heappush(pq, (ng + h[nxt], nxt, path + [nxt], ng))

path, distance = astar('Andheri Railway Station', 'Infinity Mall')

print("Optimal Path Found:")
print(" -> ".join(path))
print("\nTotal Distance =", distance, "km")

G = nx.DiGraph()

for u in graph:
    for v, w in graph[u].items():
        G.add_edge(u, v, weight=w)

plt.figure(figsize=(10,8))

nx.draw_networkx_nodes(G, pos, node_color='lightblue', node_size=2500)
nx.draw_networkx_edges(G, pos, edge_color='gray', arrows=True)

nx.draw_networkx_edges(
    G,
    pos,
    edgelist=list(zip(path, path[1:])),
    edge_color='orange',
    width=4,
    arrows=True
)

nx.draw_networkx_labels(
    G,
    pos,
    {n: f'{n}\nh(n)={h[n]}' for n in graph},
    font_size=9,
    font_weight='bold'
)

nx.draw_networkx_edge_labels(
    G,
    pos,
    {(u, v): f'{w} km' for u, v, w in G.edges.data('weight')},
    font_color='red'
)

plt.title("A* Search Route\nAndheri Railway Station to Infinity Mall")
plt.axis("off")
plt.show()
```

RBFS

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```
import networkx as nx
import matplotlib.pyplot as plt
```

```
graph = {
    'Andheri Railway Station': {'Bandra': 10, 'Jogeshwari': 4},
    'Jogeshwari': {'Bandra': 8},
    'Bandra': {'Dadar': 9},
    'Dadar': {'Byculla': 8},
    'Byculla': {'CST': 5},
    'CST': {}
}
```

```
h = {
    'Andheri Railway Station': 24,
    'Jogeshwari': 22,
    'Bandra': 14,
    'Dadar': 8,
    'Byculla': 4,
    'CST': 0
}
```

```
pos = {
    'Andheri Railway Station': (1,8),
    'Jogeshwari': (3,7),
    'Bandra': (2,5),
    'Dadar': (3,3),
    'Byculla': (4,2),
    'CST': (5,1)
}
```

```
def rbfs(node, goal, g, limit, path):
```

```
    if node == goal:
        return path, g, g
```

```
    s = [[n, g+c, max(g+c+h[n], g+h[node])]
          for n, c in graph[node].items()]
```

```
    if not s:
        return None, float('inf'), float('inf')
```

```
    while True:
        s.sort(key=lambda x: x[2])
        best = s[0]
```

```
        if best[2] > limit:
            return None, float('inf'), best[2]
```

```
        alt = s[1][2] if len(s) > 1 else float('inf')
```

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```
result, dist, best[2] = rbfs(
    best[0], goal, best[1],
    min(limit, alt),
    path + [best[0]]
)

if result:
    return result, dist, best[2]

path, distance, _ = rbfs(
    'Andheri Railway Station',
    'CST',
    0,
    float('inf'),
    ['Andheri Railway Station']
)

print("Optimal Path Found:")
print(" -> ".join(path))
print("\nTotal Distance =", distance, "km")

G = nx.DiGraph()

for u in graph:
    for v, w in graph[u].items():
        G.add_edge(u, v, weight=w)

plt.figure(figsize=(10,8))

nx.draw_networkx_nodes(G, pos, node_color='lightblue', node_size=2500)
nx.draw_networkx_edges(G, pos, edge_color='gray', arrows=True)

nx.draw_networkx_edges(
    G, pos,
    edgelist=list(zip(path, path[1:])),
    edge_color='green',
    width=4,
    arrows=True
)

nx.draw_networkx_labels(
    G, pos,
    {n: f'{n}\nh(n)={h[n]}' for n in graph},
    font_size=9,
    font_weight='bold'
)

nx.draw_networkx_edge_labels(
    G, pos,
    {(u,v): f'{w} km' for u,v,w in G.edges.data("weight")},
    font_color='red'
)
```

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```
plt.title("RBFS Route\nAndheri Railway Station to CST")  
plt.axis("off")  
plt.show()
```

OUTPUT:

```
Optimal Path Found:  
Andheri Railway Station -> DN Nagar Metro -> Lokhandwala Circle -> Infinity Mall  
Total Distance = 7 km
```

